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Human–GenAI collaboration across creative phases: Cognitive mechanisms shaping novelty and usefulness

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ABSTRACT

Generative artificial intelligence (GenAI) has demonstrated remarkable potential in creative domains; however, existing research has yet to fully explore how to optimize human–GenAI synergy or uncover the mechanisms underlying its impact on creativity. This study transcends the view of creativity as a singular process or monolithic outcome and examines how human–GenAI collaboration differentially influences the distinct creative phases of idea generation and idea elaboration. Drawing on creative process theory and goal orientation theory, we propose that human–GenAI collaboration activates phase-specific psychological mechanisms in pursuit of divergent creative goals. Using a multimethod approach, including a qualitative study (N = 20), two lab experiments (N_1 = 42; N_2 = 44), and an online behavioral experiment (N = 198), we find that human–GenAI collaboration in the idea generation phase (compared with the elaboration phase) enhances creative novelty by increasing cognitive flexibility, whereas human–GenAI collaboration in the idea elaboration phase (compared with the generation phase) improves creative usefulness by reducing cognitive overload. Furthermore, perceived GenAI intelligence moderates the indirect relationship between collaboration phases and usefulness through cognitive overload. By deconstructing the creative process, this research offers a nuanced understanding of the mechanisms and boundary conditions of human–GenAI collaboration, advancing theoretical conversations on AI-assisted creativity and providing actionable guidance for integrating GenAI into creative and organizational workflows.

1. Introduction

Since the release of ChatGPT in November 2022, generative AI (GenAI) tools such as ChatGPT, DeepSeek, and Midjourney have surged in popularity and evolved at an unprecedented pace (Dwivedi et al., 2024; Epstein et al., 2023). In the McKinsey Global Survey on AI, 65 % of respondents reported that their organizations have adopted GenAI (Singla et al., 2024). By identifying latent patterns in large datasets, GenAI fundamentally augments human creativity through its advanced content-generation capabilities (Benbya et al., 2024; Boussioux et al., 2024; Dwivedi et al., 2023; Riemer & Peter, 2024). It offers transformative advantages and reshapes workflows across domains such as music composition, digital art, and narrative design (Kshetri et al., 2024; Nedungadi et al., 2024; Zhou & Lee, 2024). A notable example is Jason Allen's AI-generated artwork *Théâtre D'opéra Spatial*, which won first prize at the Colorado State Fair.

Human–GenAI collaboration refers to the cooperative interactions between humans and generative AI systems designed to achieve collective intelligence in creative tasks (Kong et al., 2023). Although existing studies have established the positive effects of human–GenAI collaboration on individual creative performance (Doshi & Hauser, 2024; Raisch & Fomina, 2025), critical gaps remain in our understanding of this emerging phenomenon. Compared with the question of whether human–GenAI collaboration is beneficial for creativity, less attention has been given to how and when human and AI cooperation is more efficient in creative work (Flavián et al., 2024; Gehman et al., 2025). Specifically, previous studies on human–GenAI collaboration have focused on creativity as a single, linear outcome while overlooking its progression through specific creative phases (Perry-Smith & Mannucci, 2017; Zhao et al., 2024; Zhou & Lee, 2024). However, the use of GenAI to fulfill the needs of real creative work tends to be nuanced and diverse rather than uniform (Lee & Chung, 2024). Since the functions

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and influences of human–GenAI collaboration may differ across distinct creative stages, studying processual variation by associating it with different cognitive pathways and creativity-related outcomes is essential for organizations to adopt GenAI more effectively (Belanche et al., 2024).

To address this issue, we adopt a phase-specific perspective that moves beyond linear input–output models to elaborate the role that human–GenAI collaboration plays in creative work. Drawing on creative process theory (Perry-Smith & Mannucci, 2017) and goal orientation theory (Elliot & Church, 1997), we focus on the idea generation and elaboration phases and examine how human–GenAI collaboration influences two key dimensions of creativity (novelty and usefulness) by activating distinct cognitive mechanisms aligned with phase-specific goals (Mannucci & Perry-Smith, 2022). Specifically, we posit that each creative phase is guided by a distinct goal orientation: idea generation emphasizes novelty, requiring exploratory and divergent thinking (Mannucci & Perry-Smith, 2022), whereas idea elaboration prioritizes usefulness, demanding refinement and contextual integration (Mumford, 2000). The collaboration between humans and GenAI assumes different roles depending on the varying goals at different phases. In the idea generation phase, the use of GenAI supports the goal of obtaining novel solutions by reducing self-anchoring and promoting cognitive flexibility. In contrast, in the elaboration phase, the use of GenAI contributes to the goal of usefulness by converging information and reducing cognitive overload (Pezzulo et al., 2024). Moreover, compared with traditional AI, GenAI is highly dependent on input and interaction (Lin, 2024); that is, the user crucially affects the output. Guided by goal orientation theory, we posit that users' perceptions of GenAI intelligence and their prior GenAI experience can be two critical moderators that influence how effectively GenAI is used to pursue novelty or usefulness across different creative phases (Dinger et al., 2013; Gao et al., 2025).

This study addresses three research objectives: (1) identify varying goals at different creative phases and explore the phase-specific influences of human–GenAI collaboration on creativity; (2) examine the comparative influences and cognitive mechanisms of the human–GenAI collaboration phases on dual creativity outcomes; and (3) assess the moderating role of perceived GenAI intelligence and GenAI experience on these phase-contingent relationships. To test our framework, we employed a multimethod design including a qualitative study ($N = 20$), two lab experiments ($N_1 = 42$; $N_2 = 44$), and an online behavioral study ($N = 198$). The research model is presented in Fig. 1.

This research makes three core theoretical contributions to the literature on human–AI collaboration and creativity. First, we introduce a phase-goal alignment framework to advance the understanding of AI-assisted creativity (Boussioux et al., 2024; Doshi & Hauser, 2024). By unpacking how human–GenAI collaboration aligns with phase-specific creative goals (i.e., novelty in the idea generation phase and usefulness in the idea elaboration phase), this study reveals the dynamic nature of GenAI usage in fulfilling diverse work needs. Second, we respond to recent calls to explore how human–GenAI collaboration shapes human cognitive and creative capacities during work by investigating cognitive flexibility and overload as the two mechanisms (Belanche et al., 2024). Third, we also contribute by focusing on perceived GenAI intelligence and prior GenAI experience as moderators of the effectiveness of phase-specific collaboration. Identifying which types of employees are more suitable and capable of collaborating with GenAI will help managers implement more targeted and effective management strategies.

2. Theory and hypotheses

2.1. Integrating creative process theory and goal orientation theory

According to creative process theory (Perry-Smith & Mannucci, 2017), the creative process consists of four phases: idea generation, idea elaboration, idea championing, and idea implementation. This study focuses on idea generation and elaboration, which are widely considered the most critical stages and are especially well suited for human–AI collaboration (Mannucci & Perry-Smith, 2022; Runco & Jaeger, 2012). Idea generation involves the production of novel and valuable ideas through associative thinking and variation (Mumford et al., 2012; Perry-Smith & Mannucci, 2017). It emphasizes breaking away from conventional frameworks to generate unprecedented concepts (Amabile, 1983) and is achieved through divergent exploration that transcends existing practices (Mannucci & Perry-Smith, 2022; Perry-Smith & Mannucci, 2017). In contrast, idea elaboration entails systematically evaluating novel ideas, refining their value, and reducing uncertainty through iterative development (Perry-Smith & Mannucci, 2017). This phase requires balancing innovation risks with cognitive tension (Criscuolo et al., 2014; Mainemelis, 2010). Under greater cognitive load and pressure, individuals focus on translating ideas into feasible solutions while adopting more cautious strategies to mitigate failure.

According to goal orientation theory (Dweck, 1986; Han et al., 2025; Nicholls, 1984), goal orientation is a context-dependent behavioral tendency that shapes how individuals interpret tasks and regulate their cognitive, emotional, and behavioral responses. For example, tasks that prioritize novelty and openness usually activate approach-oriented goals and foster proactive exploration strategies (Miron-Spektor et al., 2022). Conversely, tasks emphasizing usefulness and risk control usually elicit avoidance-oriented goals and risk-averse behavior (Miron-Spektor et al., 2022; Simmons & Ren, 2009). As each creative phase imposes distinct demands, individuals adopt phase-specific goal orientations accordingly

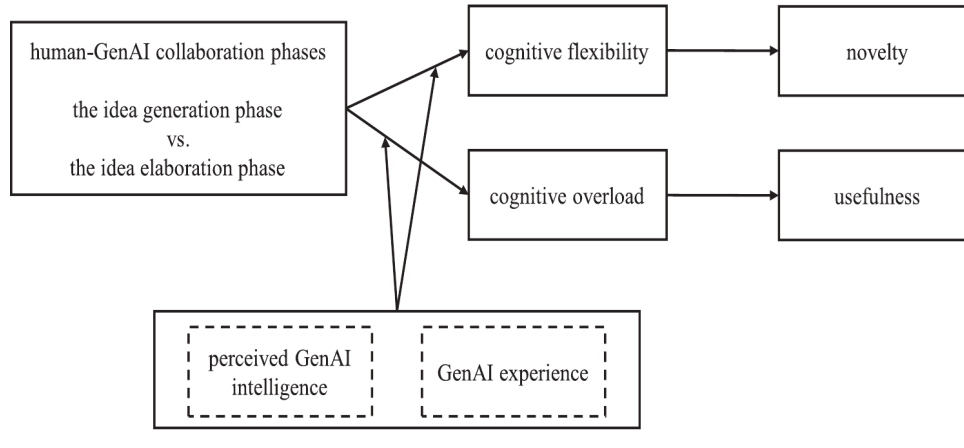


Fig. 1. Research model.

(Elliot et al., 2011). During the idea generation phase, individuals pursue novel solutions by using divergent thinking to generate unconventional ideas through broad associative exploration (Acar & Runco, 2019; Wadinambiarachchi et al., 2024). The idea elaboration phase involves convergent thinking—a focused cognitive process characterized by narrowing, evaluating, and refining ideas (Dechaume et al., 2024). During this phase, individuals adjust their goals to manage the cognitive strain caused by the heightened demands of trade-offs.

2.2. Influences of human–GenAI collaboration across different creative phases

2.2.1. Phase-specific influences of human–GenAI collaboration on dual creativity outcomes

Creativity is commonly defined as the production of ideas that are both novel and useful (Amabile, 1983; Chen & Chan, 2024; Runco & Jaeger, 2012). Achieving novelty requires divergent thinking to explore a wide range of possibilities, whereas usefulness demands convergent thinking to evaluate and refine ideas (Perry-Smith & Mannucci, 2017). In the creative process, individuals alternate between these goals and engage in distinct cognitive strategies across different phases. We argue that human–GenAI collaboration can enhance creative outcomes through alignment with these phase-specific demands.

Cognitive flexibility, defined as the capacity to break habitual thinking patterns and approach problems from multiple angles (Guilford, 1967), is critical for the development of novelty in the idea generation phase (Kundro, 2023; Perry-Smith & Mannucci, 2017). Recent research highlights that such flexibility is context sensitive and can be activated by goal structures and environmental cues (Keem et al., 2018; Kundro, 2023; Sassenberg et al., 2022). For example, work moralization can foster flexible cognition that enhances creativity (Kundro, 2023). Following this notion, we treat cognitive flexibility as a malleable state that can be shaped by human–GenAI collaboration.

In the absence of external stimuli, individuals tend to rely on pre-existing cognitive frameworks, thereby limiting the scope of their ideation (Amabile et al., 1996; Dane, 2010). GenAI mitigates this tendency by providing associative prompts that facilitate semantic expansion and reveal cross-category connections (Chowdhury et al., 2023; Criscuolo et al., 2014). This fosters cross-domain knowledge integration (Mueller & Kamdar, 2011) and encourages unconventional alternatives—a key marker of cognitive flexibility (Keem et al., 2018; Martin & Rubin, 1995). Furthermore, the neutral stance of GenAI reduces interpersonal pressure and self-censorship, allowing individuals to explore uncertain or unconventional possibilities (Siemon, 2022). Therefore, collaborating more with GenAI helps enhance cognitive flexibility, which in turn fosters creative novelty (Baas et al., 2011; Beghetto & Kaufman, 2007; Gilhooly & Murphy, 2005; Zhang et al., 2018). Thus, we propose Hypothesis 1a.

H1a. In the idea generation phase, human–GenAI collaboration enhances creative novelty by promoting cognitive flexibility.

In the idea elaboration phase, the focus shifts toward the usefulness of ideas and whether the outcome is logical and achievable (Criscuolo et al., 2014; Mumford, 2000). At this stage, individuals must mobilize cognitive resources to consider feasibility, contextual constraints, and risks when refining ideas (Perry-Smith & Mannucci, 2017; Zahra et al., 2020). These demands heighten the likelihood of cognitive overload, a state in which cognitive processing demands exceed an individual's capacity, resulting in diminished attention and reduced evaluative depth (Burmeister et al., 2022; da Silva Cezar & Maçada, 2023; Zahra et al., 2020).

Human–GenAI collaboration plays a pivotal role in mitigating this overload. By offering constructive and actionable suggestions, GenAI helps individuals reallocate mental effort more efficiently and focus on higher-order refinement tasks (Grand et al., 2016; Harrison & Rouse, 2015). In iterative development processes, GenAI can assist in identifying logic gaps, improving structural coherence, and clarifying the contextual relevance of ideas (Fui-Hoon Nah et al., 2023; Siemon, 2022). This structured support reduces unnecessary processing burdens and promotes more focused and effective elaboration. Thus, we argue that the greater the degree of cooperation with GenAI is, the less likely employees are to experience cognitive overload during the idea elaboration process, and the more likely they are to generate useful results.

H1b. In the idea elaboration phase, human–GenAI collaboration enhances creative usefulness by reducing cognitive overload.

2.2.2. Comparative influences of the human–GenAI collaboration phases on dual creativity outcomes and their mechanisms

To further explore how human–GenAI collaboration aligns with phase-specific creative goals and cognitive demands, we compare the effects of human–GenAI collaboration on creativity outcomes and the underlying psychological mechanisms in the two creative phases.

According to goal orientation theory (Dweck, 1986; Nicholls, 1984), individuals in the idea generation phase are naturally guided by novelty-seeking goals. These goals motivate them to embrace challenges, think proactively, and explore broadly through flexible cognition (Boyacı et al., 2024; Elliot et al., 2011). In this phase, creators collaborate with GenAI to engage in broad associative thinking to generate diverse ideas and select those with the greatest novelty potential (Perry-Smith & Mannucci, 2017; Mannucci & Perry-Smith, 2022). GenAI usage enables cognitive flexibility by introducing cross-domain, multiperspective alternatives that disrupt habitual thinking and reveal previously unrecognized associations (Chowdhury et al., 2023; Förster & Dannenberg, 2010). In contrast, during the idea elaboration phase, creators tend to engage in a goal-constrained, evaluative process that focuses on refining an already selected idea (Perry-Smith & Mannucci, 2017). Their collaboration with GenAI will also place greater emphasis on the pursuit of this goal. Therefore, the idea generation phase offers a more suitable condition for GenAI to activate cognitive flexibility and increase the likelihood of generating novel creative outcomes. Thus, we propose Hypotheses 2 and 3.

H2: Individuals exhibit greater cognitive flexibility during the generation phase of human–GenAI collaboration than during the elaboration phase.

H3: Cognitive flexibility mediates the effect of the human–GenAI collaboration phase (idea generation phase vs. idea elaboration phase) on creative novelty. Human–GenAI collaboration in the idea generation phase (vs. the idea elaboration phase) leads individuals to exhibit greater cognitive flexibility and thus achieve greater creative novelty.

Usefulness refers to the practical applicability and contextual relevance of ideas to specific problems or goals (Amabile, 1983). It is closely aligned with the goal orientation of the elaboration phase, which emphasizes solution refinement and implementation (Dweck, 1986; Perry-Smith & Mannucci, 2017). During the idea elaboration phase, the need for systematic evaluation aiming to minimize risks and uncertainties becomes salient (Mannucci & Perry-Smith, 2022; Perry-Smith & Mannucci, 2017). GenAI offers structured support during this phase by guiding users through in-depth analysis, identification of logical gaps, and iterative refinement of ideas. In contrast, the idea generation phase is characterized by openness and exploration (Mannucci & Perry-Smith, 2022; Perry-Smith & Mannucci, 2017). Reducing the mental burden and achieving usefulness are not the main focuses of this stage. Therefore, compared with the idea generation phase, human–GenAI collaboration during the elaboration phase is better suited to reducing cognitive overload and improving creative usefulness. Thus, we propose Hypotheses 4 and 5.

H4: Individuals exhibit lower cognitive overload during the elaboration phase of human–GenAI collaboration than during the generation phase.

H5: Cognitive overload mediates the effect of the human–GenAI collaboration phase (idea generation phase vs. idea elaboration phase) on creative usefulness. Human–GenAI collaboration in the idea elaboration phase (vs. the idea generation phase) leads individuals to exhibit

lower cognitive overload and thus achieve greater creative usefulness.

2.2.3. Moderating role of perceived GenAI intelligence and GenAI experience

As a language-based system, GenAI produces outputs that are highly user-dependent (Lin, 2024). The quality of idea generation and elaboration support GenAI tools provide hinges on how clearly individuals articulate their goals and guide the responses of the GenAI tool (White et al., 2023; Zamfirescu-Pereira et al., 2023). According to goal orientation theory, individuals' perceptions of their collaborator's competence and their own alignment with task demands influence the extent to which external support enhances goal pursuit (Dinger et al., 2013; Duong et al., 2023; Gao et al., 2025). Perceived GenAI intelligence refers to the extent to which users view GenAI as a capable and trustworthy collaborator (Chen et al., 2023; Warner and Sugarman, 1986), whereas GenAI experience, often measured by the frequency of prior use, reflects users' ability to formulate targeted prompts and navigate GenAI responses effectively (Chowdhury et al., 2022; Man Tang et al., 2022). We argue that both factors shape the comparative utility of human–GenAI collaboration by strengthening an individual's ability to align GenAI's contribution with phase-specific creative goals.

GenAI supports the idea generation process by offering associative cues, cross-domain inputs, and idea-disruptive stimuli (Chowdhury et al., 2023; Mueller & Kamdar, 2011). We argue that the effectiveness of this support depends on the attitude and skill of the user. Specifically, individuals who perceive GenAI as highly intelligent are more motivated to consider its suggestions (Lalot & Bertram, 2025), and those who are experienced are better at extracting generative value and steering the interaction in a novel direction (Flavián et al., 2022; Wirtz et al., 2018). In contrast, in the idea elaboration phase, goal-constrained evaluation and early commitment to selected ideas often limit individuals' openness to divergent input (Perry-Smith & Mannucci, 2017; Rhue, 2023). GenAI-generated novel suggestions are more likely to be dismissed when they deviate from the intended refinement trajectory. Under such conditions, even individuals with strong motivation to use GenAI or extensive prior experience may find it difficult to engage in flexible exploration. As a result, GenAI input is less likely to contribute meaningfully to creative novelty during this phase. Thus, we propose Hypothesis 6.

H6a. Perceived GenAI intelligence moderates the indirect effect of the human–GenAI collaboration phase on creative novelty through cognitive flexibility such that for individuals with high (vs. low) perceived GenAI intelligence, human–GenAI collaboration during the idea generation phase (vs. the idea elaboration phase) results in greater cognitive flexibility and thus greater creative novelty.

H6b. GenAI experience moderates the indirect effect of the human–GenAI collaboration phase on creative novelty through cognitive flexibility such that for individuals with high (vs. low) GenAI experience, human–GenAI collaboration during the idea generation phase (vs. the idea elaboration phase) results in greater cognitive flexibility and thus greater creative novelty.

GenAI enhances creative usefulness in the idea elaboration phase by offering structured feedback, checking for logical coherence, and suggesting optimization strategies (Grand et al., 2016; Lee & Chung, 2024). We argue that both individuals' motivational engagement (i.e., willingness to operationalize GenAI feedback) and their prior digital experience are critical in determining how effectively GenAI reduces overload and enhances elaborative usefulness. Individuals with higher perceived GenAI intelligence are more likely to trust and consider GenAI's evaluative input, whereas those with greater GenAI experience are better equipped to frame precise prompts, select relevant suggestions, and synthesize feedback into contextually appropriate elaborations (Boussioux et al., 2024; Chen & Chan, 2024; Liu et al., 2024; Man Tang et al., 2022). These creators are thus more capable of reducing cognitive overload and achieving higher-quality refinement. In contrast, in the idea generation phase, creators typically operate without a defined creative framework. Ideas remain fluid and provisional, and the primary goal is exploratory rather than analytical. In this unstructured context, creators often lack the specificity needed to solicit meaningful elaborative feedback from GenAI or to identify actionable optimization cues. The ability of GenAI to reduce overload or enhance creative usefulness will be limited. Thus, we propose Hypothesis 7.

H7a. Perceived GenAI intelligence moderates the indirect effect of the human–GenAI collaboration phase on creative usefulness through cognitive overload; for individuals with high (vs. low) perceived GenAI intelligence, human–GenAI collaboration during the idea elaboration phase (vs. the idea generation phase) results in lower cognitive overload and thus greater creative usefulness.

H7b. GenAI experience moderates the indirect effect of the human–GenAI collaboration phase on creative usefulness through cognitive overload; for individuals with high (vs. low) GenAI experience, human–GenAI collaboration during the idea elaboration phase (vs. the idea generation phase) leads to lower cognitive overload and thus greater creative usefulness.

3. Overview of studies

We used a multimethod approach comprising a qualitative study, two laboratory experiments, and an online behavioral experiment to test the hypotheses. Specifically, Study 1 was a qualitative investigation that preliminarily explored how human–GenAI collaboration enhances individual creativity across creative phases and examined the underlying psychological mechanisms. Study 2 included two laboratory experiments designed to test Hypothesis 1 and to evaluate whether and how human–GenAI collaboration during the idea generation and elaboration phases influences creative outcomes. Study 3 was an online behavioral experiment aimed at testing Hypotheses 2 through 7, with a focus on the comparative influences, mechanisms, and boundary conditions of human–GenAI collaboration across distinct creative phases.

4. Study 1

4.1. Sample, design, and procedures

To ensure that the interviewees had prior experience with human–GenAI collaboration in creative work, the participants were recruited and screened via online advertisements and snowball sampling. Specifically, we selected professionals in creative roles who had substantial experience integrating GenAI into their work, thereby ensuring the relevance and credibility of the sample. To maintain data quality, participants were paid 60 CNY per hour. Data collection was concluded when the responses from the last five participants became highly repetitive, offering no new or significantly distinct insights, thereby reaching data saturation. A total of 20 interviews were conducted, meeting common qualitative sampling thresholds (Guest et al., 2006; Malik et al., 2023). The participants came from diverse cities and industries, with an average age of 27.8 years ($SD = 3.16$). Among them, 40 % were male, 55 % held bachelor's degrees, and 35 % had master's degrees or above (see Appendix A).

We collected data for this study through semistructured interviews. At the beginning of each session, the participants were provided with a definition of GenAI to ensure shared understanding. The interviews covered four key areas. First, the participants reported their basic demographic information and GenAI usage details, including duration and the specific tools employed. Second, they described how they collaborated with GenAI, offering examples of the creative phases in which such collaboration occurred. Third, they articulated their goals when using GenAI for creative tasks. Finally, they reflected on the value of collaborating with GenAI, detailing its tangible impacts on their creative work and sharing personal insights from their experiences (see Appendix B).

4.2. Data analysis

With participants' consent, all interviews were audio-recorded, with each session lasting approximately 50 min on average. The recordings were transcribed into 139,000 words of textual data. We used NVivo 12 for manual coding and followed a procedural grounded theory framework comprising three stages: open, axial, and selective coding. During open coding, we identified 25 initial concepts. These were then systematically compared and categorized during axial coding, resulting in six subcategories. Finally, in the selective coding stage, we examined interrelationships among subcategories and synthesized them into three core categories.

The interview data revealed that creative professionals primarily engaged in human–GenAI collaboration during the idea generation and elaboration phases. For example, one participant shared their use of GenAI to spark inspiration during idea generation: "Sometimes when I'm out of ideas, I let GenAI help brainstorm some plot points for inspiration." Another participant described employing GenAI during elaboration: "I have AI analyze the advantages and disadvantages of different plans and propose adding surprise elements, raffles, and other refinements." These findings highlight distinct phases of human–GenAI collaboration, consolidated into the core category of *the human–GenAI collaboration phases*.

The participants' collaborative goals varied. Some prioritized creative novelty, as illustrated by one participant: "I pay attention to novelty and consider how to grab more attention." Others focused on creative usefulness, such as ensuring practicality and relevance: "I pay attention to whether it's realistic, for example, whether a raffle giveaway of clothes might lead to product complaints." These contrasting goals were categorized under *the creative goals of the human–GenAI collaboration* .

Additionally, the participants identified key benefits of collaborating with GenAI, including enhanced cognitive flexibility: "GenAI designed kites with unconventional, fun imagery that defies traditional styles, creating innovative ideas that break away from the norm." Others emphasized its role in alleviating creative pressure: "GenAI is like a reliable colleague who always has my back, never lets me down, and is always ready to help." These observations reflected the psychological mechanisms by which GenAI shaped creative work and were grouped according to *underlying psychological mechanisms*.

An in-depth analysis revealed that participants who collaborated with GenAI during the idea generation phase tended to prioritize novelty, often aiming to capture attention through distinctive and original ideas. In contrast, those engaging with GenAI during the elaboration phase often aimed to enhance creative usefulness by reducing risks and improving feasibility. The main categories, subcategories, initial concepts, and illustrative quotes from Study 1 are presented in Appendix C (see Appendix C).

Through semistructured interviews, this study preliminarily validated its research framework, confirming that human–GenAI collaboration in creative work was phase dependent. These findings offered an initial foundation for understanding the complex dynamics of the human–GenAI interaction within the creative process. However, while the qualitative approach yielded valuable insights, it was inherently limited in scientific rigor and generalizability. To address these limitations, the next phase employed an experimental methodology incorporating quantitative techniques to further test the proposed research model. By systematically analyzing the relationships and mechanisms identified during the qualitative phase, this quantitative study aimed to provide stronger empirical evidence, thereby enhancing the scientific rigor and generalizability of the findings. By integrating qualitative and quantitative approaches, this research adopted a more holistic and balanced investigative framework, ultimately deepening our understanding of human–GenAI collaboration in creative contexts.

5. Study 2

We conducted two laboratory experiments in Study 2 to enable strict experimental control, which was essential for precisely manipulating human–GenAI collaboration (or its absence) across different creative phases.

5.1. Study 2a

5.1.1. Sample, design, and procedures

Study 2a employed a within-subjects design, comparing collaboration with GenAI versus working independently during the idea generation phase. A total of 45 university students were recruited through online and offline campus groups, with each participant receiving 40 CNY upon completion. Following Meade and Craig's (2012) recommendations, an attention check item was included to detect careless responses (i.e., participants were instructed to select "strongly disagree" for the item). Three participants failed the check, resulting in a final sample of 42. The gender distribution was nearly balanced, with 35.7 % identifying as male. The average age was 21.17 years (SD = 2.25); 59.5 % held bachelor's degrees, 33.3 % held master's degrees, and 7.1 % were pursuing doctoral degrees.

Before the experiment, the participants received instructions outlining the procedure, task requirements, and use of the "SparkDesk" tool. SparkDesk is a Chinese-developed cognitive intelligence large language model (LLM) with core capabilities that include text generation, knowledge-based question answering, and logical reasoning (Cheng et al., 2023). Each participant completed two product marketing-related creativity tasks (Task A and Task B) in sequence. A key distinction lies in task execution: in one task, participants collaborated with GenAI during idea generation; in the other, they worked independently. Post-task measures included scales assessing cognitive flexibility and cognitive overload, along with demographic information (gender, age, and educational level). Research assistants monitored the task duration, with each session lasting approximately 20 min.

The creative task in Study 2a was adapted from Noy and Zhang (2023) (see Appendix D), and the descriptive materials for the idea generation and elaboration phases were drawn from Mannucci and Perry-Smith (2022) (see Appendix E). To address potential learning transfer effects and mitigate differences in task difficulty, we employed two product marketing tasks of equivalent difficulty. Prior to the experiment, five Ph.D. students specializing in human resources and marketing evaluated the tasks to ensure difficulty equivalence. During the experiment, a measurement item was embedded (e.g., "How different in difficulty do you perceive the two completed tasks to be?"). Both the task order and the human–GenAI collaboration order were randomized to control for order effects.

5.1.2. Measures

Unless noted, all the measures were assessed on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

Cognitive Flexibility. To measure cognitive flexibility, we used three items from Martin and Rubin (1995). A sample item was "When dealing with the creative task, I consider alternatives" ($\alpha_{\text{H condition}} = 0.70$, $\alpha_{\text{GenAI condition}} = 0.76$; the H condition was the condition in which participants complete the task independently, and the GenAI condition was the condition in which participants collaborate with GenAI in the idea generation phase).

Creativity. Referring to previous studies (e.g., Man Tang et al., 2022; Zhang et al., 2022), we used the consensual assessment technique to assess creativity (Amabile, 1983). Specifically, we invited three experts from the field of marketing to evaluate the participants' creative proposals without knowing the purpose, hypotheses, or conditions of the experiment. They rated the proposals on the basis of whether "the proposal had a high degree of usefulness" and whether "the proposal had a high degree of novelty." The definition of usefulness (50 %) was

whether the proposal had the potential for practical implementation and whether it could solve real problems and create maximum value. The definition of novelty (50 %) was the degree of innovation and uniqueness of the proposal and advertising copy. Owing to the high consistency among the three experts in evaluating the proposals of each participant in terms of novelty (ICC2 $_{\text{H condition}} = 0.82$, $p < 0.01$; ICC2 $_{\text{GenAI condition}} = 0.85$, $p < 0.001$) and usefulness (ICC2 $_{\text{H condition}} = 0.87$, $p < 0.01$; ICC2 $_{\text{GenAI condition}} = 0.82$, $p < 0.001$), the average score from the three experts was taken as the final score for each participant.

Control variables. We controlled for the participants' age, gender (0 = male, 1 = female), and education background. Previous studies have shown that these demographic variables can influence creativity (e.g., Jeong et al., 2023; Madjar et al., 2011; Tang & Naumann, 2016; Tierney & Farmer, 2002; Zhang et al., 2022).

5.1.3. Results

Preliminary analysis. To avoid the influence of difficulty differences between two tasks on the creativity scores, we measured the perceived difficulty difference between the two tasks by the participants. The item was "How much do you think the two tasks you just completed differ in terms of difficulty?" The measure was assessed on a 7-point Likert scale ranging from 1 (no difference) to 7 (a large difference). The results of the one-sample t test revealed that the mean score was significantly less than 4 ($M = 2.36$, $SD = 0.85$, $t = -12.52$, $p < 0.001$), indicating that the two tasks were perceived as having little practical difference in difficulty. Table 1 shows the results, including the means, standard deviations, and correlations among the study variables.

Hypothesis testing. We used SPSS software to estimate the indirect effect of human–GenAI collaboration on novelty through cognitive flexibility in the idea generation phase (Hayes, 2017; Montoya & Hayes, 2017). On the basis of 5000 bootstrapping replications, the indirect effect of human–GenAI collaboration in the idea generation phase on novelty through cognitive flexibility was significant (estimate = 0.17, 95 % CI = [0.033, 0.341]), thereby supporting Hypothesis 1a.

5.2. 5.2. Study 2b

5.2.1. Sample, design, and procedures

Approximately one month after Study 2a, we conducted Study 2b. There was no sample overlap, as participants were recruited separately to preserve the integrity of the research design. As in Study 2a, Study 2b adopted a within-subjects experimental design, manipulating whether participants collaborated with GenAI during the idea elaboration phase. Using the same recruitment methodology as in Study 2a, we enrolled 45 university students. One participant failed the attention check, resulting in a final sample of 44. The gender distribution was nearly balanced, with 34.1 % identifying as male. The average age was 21.16 years ($SD = 2.13$); 63.6 % held bachelor's degrees, 34.1 % held master's degrees, and 2.3 % were pursuing doctoral degrees.

The participants completed two product marketing-related creativity tasks in sequence (see Appendix D). A key distinction lay in task execution: in one task, participants collaborated with GenAI during the idea elaboration phase, whereas in the other, they worked independently (see Appendix F). The randomization procedures in Study 2b were identical to those used in Study 2a.

5.2.2. Measures

Cognitive Overload. To measure cognitive overload, we used three items from Hart (1988) and Sung et al. (2017). A sample item was "I found that task to be very mentally demanding" ($\alpha_{\text{H condition}} = 0.89$, $\alpha_{\text{GenAI condition}} = 0.82$; the H condition was the condition in which participants completed the task independently, and the GenAI condition was the condition in which participants collaborated with GenAI in the idea elaboration phase).

Creativity. To measure participants' creativity, we used the same measurement used in Study 2a (novelty: ICC2 $_{\text{H condition}} = 0.79$, $p < 0.01$; ICC2 $_{\text{GenAI condition}} = 0.74$, $p < 0.001$; usefulness: ICC2 $_{\text{H condition}} = 0.91$, $p < 0.01$; ICC2 $_{\text{GenAI condition}} = 0.93$, $p < 0.001$).

Control variables. Consistent with Study 2a, we controlled for the participants' age, gender (0 = male, 1 = female), and education background.

5.2.3. Results

Preliminary analysis. Consistent with Study 2a, we measured the difficulty level difference between the two tasks as perceived by the participants. The results of the one-sample t test revealed that the mean score was significantly less than 4 ($M = 2.32$, $SD = 0.86$, $t = -13.03$, $p < 0.001$), indicating that the two tasks were perceived as having little practical difference in difficulty. Table 2 shows the results, including the means, standard deviations, and correlations among the study variables.

Hypothesis testing. We also used SPSS software to estimate the indirect effect of human–GenAI collaboration on usefulness through cognitive overload during the idea elaboration phase. On the basis of 5000 bootstrapping replications, the indirect effect of human–GenAI collaboration in the idea elaboration phase on usefulness through cognitive overload was significant (estimate = 0.14, 95 % CI = [0.056, 0.222]), thus supporting Hypothesis 1b.

6. Study 3

6.1. Sample, design, and procedures

We conducted an online behavioral experiment in Study 3 to examine the influences, mediating mechanisms, and boundary

Table 1
Correlation coefficients for all variables in Study 2a.

Variables	M	SD	1	2	3	4	5	6	7	8	9	10	11	12
1.Gender	0.64	0.49												
2.Age	21.17	2.25	0.17											
3.Education	5.48	0.63	0.25	0.90**										
4.CO (H condition)	5.14	0.93	- 0.16	- 0.06	- 0.16									
5.CO (GenAI condition)	4.13	1.31	0.18	- 0.01	0.01	0.45**								
6.CF (H condition)	5.15	1.05	0.01	0.11	0.10	- 0.04	- 0.20							
7.CF (GenAI condition)	5.48	0.97	- 0.01	0.30	0.25	- 0.11	- 0.39*	0.67**						
8.Novelty (H condition)	5.44	0.55	- 0.09	- 0.07	- 0.11	0.11	0.15	0.07	- 0.29					
9.Novelty (GenAI condition)	6.05	0.59	0.26	0.13	0.16	- 0.29	- 0.15	- 0.25	- 0.02	0.14				
10.USEN (H condition)	5.21	0.45	- 0.04	0.02	- 0.05	0.33*	0.10	- 0.05	- 0.06	0.43**	0.05			
11.USEN (GenAI condition)	5.37	0.52	0.08	0.11	0.08	- 0.15	- 0.23	- 0.31**	0.06	- 0.02	0.61**	- 0.06		
12.GenAIE	3.43	1.53	- 0.02	0.01	0.01	- 0.02	- 0.01	0.08	0.05	0.06	- 0.11	0.23	- 0.04	
13.PGenAII	5.58	0.74	- 0.16	- 0.06	- 0.03	- 0.02	- 0.27	0.30	0.42**	0.01	0.03	0.02	- 0.01	0.08

Note: $N = 42$; SD = standard deviation.
CF (cognitive flexibility); CO (cognitive overload); USEN (usefulness); GenAIE (GenAI experience); PGenAII (perceived GenAI intelligence).
*p < .05, **p < .01.

Table 2
Correlation coefficients for all variables in Study 2b.

Variables	M	SD	1	2	3	4	5	6	7	8	9	10	11	12
1.Gender	0.66	0.48												
2.Age	21.16	2.13	0.17											
3.Education	5.39	0.54	0.07	0.84**										
4.CF (H condition)	5.03	1.09	0.02	- 0.24	- 0.35*									
5.CF (GenAI condition)	4.95	1.11	0.03	- 0.14	- 0.33*	0.76**								
6.CO (H condition)	5.19	1.45	- 0.05	0.09	0.11	0.02	- 0.08							
7.CO (GenAI condition)	4.64	1.42	0.13	0.21	0.21	- 0.07	- 0.20	0.65**						
8.USEN (H condition)	5.49	0.39	0.07	0.05	0.11	- 0.21	- 0.23	- 0.24	- 0.06					
9.USEN (GenAI condition)	5.89	0.39	- 0.09	- 0.12	- 0.02	- 0.13	- 0.05	0.34*	- 0.12	0.30				
10.Novelty (H condition)	5.31	0.47	- 0.24	- 0.15	- 0.03	0.16	0.08	- 0.08	- 0.19	0.35*	0.30*			
11.Novelty (GenAI condition)	5.43	0.49	0.05	- 0.09	0.03	- 0.07	- 0.12	0.16	- 0.07	0.23	0.36*	0.39**		
12.GenAIE	3.42	1.50	0.04	0.15	0.20	- 0.12	0.02	0.34*	0.06	- 0.15	0.26	0.11	0.20	
13.PGenAII	5.76	0.77	- 0.20	- 0.26	- 0.34*	0.36*	0.43**	0.06	- 0.31*	- 0.28	0.17	0.15	0.25	0.05

Note: N = 44; SD = standard deviation.
CF (cognitive flexibility); CO (cognitive overload); GenAIE (GenAI experience); PGenAII (perceived GenAI intelligence).
*p < .05, **p < .01.

conditions of human–GenAI collaboration phases (idea generation vs. idea elaboration) on two creativity outcomes. Study 3 adopted a 2 (creative phase: idea generation vs. idea elaboration) × 2 (perceived GenAI intelligence: high vs. low) between-subjects design. Given the difficulty of experimentally manipulating GenAI experience (an additional moderator), this variable was measured using self-reported scales. To standardize human–GenAI collaboration, we developed a custom online system that guided participants' interactions with GenAI during specific creative phases (Chen & Chan, 2024). The system integrated a GenAI interface via an application programming interface (API—a software bridge that allows different systems to communicate and exchange data) to enable controlled collaboration, with technical support provided by professional software engineers.

G*Power 3.1 (Faul et al., 2007) indicated that a two-factor ANOVA (f = 0.25, α = 0.05, four groups) requires at least 146 participants to achieve 85 % power. We recruited 205 full-time employees through purposive and snowball sampling. After seven participants who failed attention checks were excluded, 198 valid responses remained for analysis. The gender distribution was nearly balanced, with 59.1 % identifying as male. The average age was 27.60 years (*SD* = 4.00); 75.3 % held bachelor's degrees, and 19.7 % held master's degrees. With respect to industry sectors, 26.8 % of the participants worked in services, 26.3 % in information technology, and 18.7 % in manufacturing.

The participants were invited to engage with the online system and were briefed on the experimental procedures, participation requirements, and the GenAI tool. ChatGPT, a representative GenAI platform, was used in this online experiment (Cromwell et al., 2023; Kshetri et al., 2024). A randomized 2 × 2 between-subjects design assigned participants to one of four experimental conditions. Upon completing the creative task, they responded to measures assessing demographics, cognitive flexibility, cognitive overload, and additional control variables. To ensure data quality, participants received a base compensation of 20 CNY, with additional rewards offered for exceptional creative solutions. The study was preregistered on AsPredicted (https://aspredicted.org/2hsq-9sg4.pdf).

The creative task in Study 3 was adapted from Chen and Chan (2024) (see Appendix G). The participants were asked to develop an advertising plan for a mobile phone case. This product was selected because it is a common consumer good that requires no specialized industry knowledge, ensuring accessibility for online participants. At the same time, its customizable features provided sufficient flexibility for participants to express creativity. Consistent with Study 2, the materials for manipulating the human–GenAI collaboration phase were adapted from Mannucci and Perry-Smith (2022). The manipulation of perceived GenAI intelligence combined two approaches: a news-report method adapted from Yam et al. (2021) and a recall method from Maner et al. (2007) (see Appendix H). The former presented news excerpts describing GenAI intelligence levels, whereas the latter asked participants to recall and describe recent experiences of perceiving high or low GenAI intelligence, thereby enhancing the manipulation effect through the recall paradigm.

6.2. Measures

The specific measurement items used in this study are presented in Appendix I.

Cognitive Flexibility. We used the same three items from Study 2 to measure cognitive flexibility. Cronbach's alpha for this scale was 0.81.

Cognitive Overload. We used the same three items from Study 2 to measure cognitive overload. Cronbach's alpha for this scale was 0.83.

Creativity. To measure participants' creativity, we used the same measurement approach used in Study 2 (novelty: ICC2 = 0.82, $p < 0.001$; usefulness: ICC2 = 0.84, $p < 0.001$).

Perceived GenAI intelligence. We used five items adapted from Warner and Sugarman (1986) to test the manipulation of perceived GenAI intelligence (Chen et al., 2023). A sample item was "I think GenAI is intelligent". The Cronbach's alpha for this scale was 0.87.

GenAI experience. We used three items adapted from Man Tang et al. (2022) to measure how often an individual used GenAI to complete tasks in their daily life to represent their GenAI experience. A sample item was "I used GenAI to carry out most of my job functions". Cronbach's alpha for this scale was 0.89.Man

Control variables. Based on previous research, we controlled for participants' demographic characteristics, such as age, gender (0 = male, 1 = female), educational level, and industry. Additionally, we controlled for digital knowledge, creative self-efficacy, and GenAI trust. Studies have shown that these factors are closely related to the effectiveness of human–AI interactions and that of individual creativity (Shao et al., 2022; Sohn & Kwon, 2020; Yin et al., 2024). Digital knowledge was measured using a 3-item scale developed by Seo et al. (2015). A sample item was "I have the capabilities required for using GenAI". Cronbach's alpha for this scale was 0.89. Creative self-efficacy was measured using a 3-item scale developed by Tierney and Farmer (2002). A sample item was "I have confidence in my ability to solve problems creatively". The Cronbach's alpha for this scale was 0.84. GenAI-related trust was measured using a 10-item scale developed by Yang et al. (2009) and Dunn et al. (2012). A sample item was "I will discuss the challenges I encounter in my work with GenAI". The Cronbach's alpha for this scale was 0.90.

6.3. Results

6.3.1. Preliminary analysis

Descriptive statistics and correlation analyses were conducted on the

variables involved in this study using SPSS, as shown in Table 3. According to Table 3, the human–GenAI collaboration phases (1 = idea elaboration phase, 0 = idea generation phase) were negatively correlated with cognitive flexibility ($r = -0.30$, $p < 0.01$) and cognitive overload ($r = -0.35$, $p < 0.01$). Furthermore, cognitive flexibility was positively correlated with novelty ($r = 0.49$, $p < 0.01$), and cognitive overload was negatively correlated with usefulness ($r = -0.60$, $p < 0.01$).

6.3.2. Manipulation checks

First, we validated the effectiveness of manipulating the independent variable and moderator. For the independent variable, the results of independent sample t tests revealed significant differences in the manipulation index ($t(196) = 28.63$, $p < 0.001$) between the idea generation phase ($M = 6.31$, $SD = 0.89$) and the idea elaboration phase ($M = 2.00$, $SD = 1.21$). Additionally, for the moderator variable, independent sample t tests revealed significant differences in perceived GenAI intelligence between the high-GenAI intelligence group ($M = 5.53$, $SD = 0.88$) and the low-GenAI intelligence group ($M = 4.64$, $SD = 1.24$; $t(196) = 5.81$, $p < 0.001$). These results confirmed successful manipulation of both the independent variable and moderator.

6.3.3. Hypothesis testing

Hypothesis 2. proposed that individuals would exhibit greater cognitive flexibility during the generation phase of human–GenAI collaboration than during the elaboration phase. Independent sample t tests revealed significant differences in the cognitive flexibility score of the idea generation phase ($M = 5.58$, $SD = 0.87$) and that of the idea elaboration phase ($M = 5.01$, $SD = 0.94$; $t(196) = 4.41$, $p < 0.001$). Thus, Hypothesis 2 was supported.

Hypothesis 3. proposed that cognitive flexibility mediated the effect of the human–GenAI collaboration phase (idea generation phase vs. idea elaboration phase) on creative novelty. Using Process 3.3 (Model 4) in SPSS, we calculated indirect effects via 5000 bootstrap resampling. The results revealed a significant indirect effect of the human–GenAI collaboration phases (1 = idea generation phase, 0 = idea elaboration phase) on creative novelty through cognitive flexibility (indirect effect = 0.16, SE = 0.06, 95 % CI = [0.065, 0.283]). Therefore, Hypothesis 3 was supported.

Hypothesis 4. proposed that individuals would exhibit lower cognitive overload during the elaboration phase of human–GenAI collaboration than during the generation phase. Independent sample t tests confirmed this, with significant differences in the cognitive overload scores of the human–GenAI collaboration during the idea elaboration phase ($M = 4.14$, $SD = 0.99$) and that during the idea generation phase ($M = 4.97$, $SD = 1.23$; $t(196) = 5.22$, $p < 0.001$). Thus, Hypothesis 4 was supported.

Hypothesis 5. proposed that cognitive overload mediated the effect of the human–GenAI collaboration phase (idea generation phase vs. idea elaboration phase) on creative usefulness. Bootstrap-based mediation analysis revealed a significant indirect effect of human–GenAI collaboration phases (1 = idea elaboration phase, 0 = idea generation phase) on creative usefulness via cognitive overload (indirect effect = 0.35, SE = 0.08, 95 % CI = [0.205, 0.518]). Thus, Hypothesis 5 was supported.

Hypothesis 6. proposed that both perceived GenAI intelligence (H6a) and GenAI experience (H6b) would moderate the indirect effect of the human–GenAI collaboration phases on creative novelty via cognitive flexibility. Using Process 3.3 (Model 7), we tested moderated mediation via 95 % confidence intervals for the INDEX. The results revealed no significant moderated mediation effects for either perceived GenAI intelligence (INDEX = 0.14, SE = 0.08, 95 % CI = [-0.018, 0.313]) or

Table 3
Correlation coefficients for all variables in Study 3.

Variables	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1.Gender	0.41	0.49													
2.Age	27.60	4.00	0.10												
3.Education	3.16	0.49	0.15*	-0.02											
4.Industry	3.27	2.10	0.24**	-0.13	0.20**										
5.DS	5.39	1.08	-0.12	-0.22**	0.01	-0.16*									
6.GenAIT	4.67	1.07	-0.19**	-0.22**	-0.16*	-0.12	0.40**								
7.CSE	5.14	1.14	-0.14*	-0.13	-0.12	-0.06	0.38**	0.50**							
8.CP	0.49	0.50	-0.01	0.19**	-0.01	-0.10	-0.14*	-0.07	-0.15*						
9.CF	5.30	0.95	-0.17*	-0.06	-0.09	-0.06	0.44**	0.43**	0.56**	-0.30**					
10.CO	4.57	1.19	0.06	0.01	-0.04	-0.06	-0.01	0.06	0.05	-0.35**	0.26**				
11.Novelty	4.60	0.89	-0.12	0.01	-0.02	-0.08	0.22**	0.12	0.32**	-0.30**	0.49**	0.08			
12.Usefulness	4.52	0.90	-0.03	-0.01	0.03	0.06	-0.02	-0.09	-0.03	0.42**	-0.23**	-0.60**	0.13		
13.PGenAII	5.08	1.16	-0.10	0.02	-0.06	-0.03	0.11	0.15*	-0.06	0.03	-0.04	0.09	-0.07	-0.04	
14.GenAIE	5.07	1.45	-0.02	-0.18*	0.02	-0.09	0.50**	0.42**	0.34**	0.04	0.40**	-0.01	0.23**	0.08	-0.11

Note: N = 198; SD = standard deviation.
DS(digital skills); GenAIT(GenAI trust); CSE(creative self-efficacy); CP(collaboration phases); CF (cognitive flexibility); CO (cognitive overload); PGenAII (perceived GenAI intelligence); GenAIE (GenAI experience).
*p < .05, **p < .01

GenAI experience (INDEX = - 0.002, SE = 0.03, 95 % CI = [- 0.065, 0.060]). Thus, Hypothesis 6 was not supported.

Hypothesis 7. proposed that both perceived GenAI intelligence (H7a) and GenAI experience (H7b) would moderate the indirect effect of the human–GenAI collaboration phases on creative usefulness via cognitive overload. The results revealed that after controlling for all covariates, the indirect effect of human–GenAI collaboration phases on creative usefulness via cognitive overload was stronger when perceived GenAI intelligence was high (indirect effect = 0.48, SE = 0.11, 95 % CI = [0.286, 0.704]) than when it was low (indirect effect = 0.22, SE = 0.10, 95 % CI = [0.047, 0.427]). More importantly, the difference between the indirect effects of the two groups was significant (INDEX = 0.26, SE = 0.13, 95 % CI = [0.019, 0.516]), supporting Hypothesis 7a. Although our study did not propose a hypothesis regarding the moderating effect, we conducted a moderating effect analysis. The results indicated that the interaction effect between the human–GenAI collaboration phases and perceived GenAI intelligence on cognitive overload was also significant ($F(1, 194) = 5.06$, $p < 0.05$). The simple effects analysis results are presented in Fig. 2.

However, when GenAI experience served as the moderator, the INDEX of moderated mediation was nonsignificant (INDEX = 0.04, SE = 0.05, 95 % CI = [- 0.050, 0.135]). Thus, Hypothesis 7b was not supported.

7. Discussion

This study investigates how human–GenAI collaboration influences creative outcomes across different phases of the creative process. Drawing on creative process theory and goal orientation theory, we propose that the impact of human–GenAI collaboration on creativity is not uniform but rather phase contingent. Using a multimethod approach, we demonstrate three key findings. First, individuals prioritize different creative goals across phases: novelty in idea generation and usefulness in idea elaboration. Second, human–GenAI collaboration aligns with these phase-specific goals through different psychological mechanisms. Compared with human–GenAI collaboration in the idea elaboration phase, collaboration during idea generation phase has a stronger effect on novelty by enhancing cognitive flexibility. In contrast, human–GenAI collaboration more effectively improves usefulness by reducing cognitive overload in idea elaboration than in idea generation. Third, perceived GenAI intelligence moderates the indirect effect of the collaboration phase on creative usefulness but not on novelty.

The moderating effect of perceived GenAI intelligence on the relationship between human–GenAI collaboration phases and novelty was not significant, likely because of the distinct cognitive demands of the two phases. In the idea generation phase, creators pursue cognitive flexibility, which can be readily triggered by minimal GenAI cues, without requiring deep engagement with GenAI (Finke et al., 1996). In fact, perceiving GenAI as highly intelligent too early may induce over-reliance, thereby narrowing the generation of diverse ideas (Yin et al., 2025). In contrast, the elaboration phase demands cognitive effort and more structured feedback integration, making perceived GenAI intelligence more impactful in helping creators evaluate, synthesize, and improve ideas.

Additionally, the hypotheses concerning the moderating role of GenAI experience were not supported. This nonsignificant effect may reflect a ceiling phenomenon: most participants in our sample reported high-frequency use of GenAI in their work, leading to limited variance in experience levels. As GenAI becomes increasingly prevalent in workplaces, employees generally exhibit advanced proficiency in using it, thereby limiting the variability required to examine the moderation effects.

Together, these findings reinforce the importance of viewing human–GenAI collaboration as an evolving interaction shaped by the task phase, individual cognition, and system perception. Rather than treating GenAI as a static partner, effective collaboration depends on aligning its use with the cognitive priorities and constraints of each creative phase.

7.1. Theoretical contributions

Our research makes several significant contributions. First, our research advances a dynamic, phase-specific perspective on human–GenAI collaboration, thereby refining theories related to AI-assisted creativity. While prior research has highlighted the promise of GenAI in enhancing creativity, it often adopts a static, linear view of the creative process and treats creativity as a unitary outcome (Boussioux et al., 2024; Doshi & Hauser, 2024; Zhao et al., 2024). These static models of AI augmentation overlook the fact that creativity unfolds across distinct phases, each shaped by unique goal orientations and cognitive demands. By distinguishing creative processes and framing creativity as a multidimensional construct, this dual-phase perspective responds to a growing body of research that calls for a more nuanced understanding of human–GenAI collaboration and creativity (Lee & Chung, 2024; Yeo et al., 2024). Our research expands existing resource-based and motivation-centric frameworks (Jia et al., 2024; Shao et al., 2024; Wang et al., 2022; Yin et al., 2024) by clarifying when and how human–GenAI collaboration aligns with evolving creative goals. It also helps prevent theoretical conflation of distinct cognitive mechanisms, a limitation that may obscure when and why GenAI helps or hinders creativity in

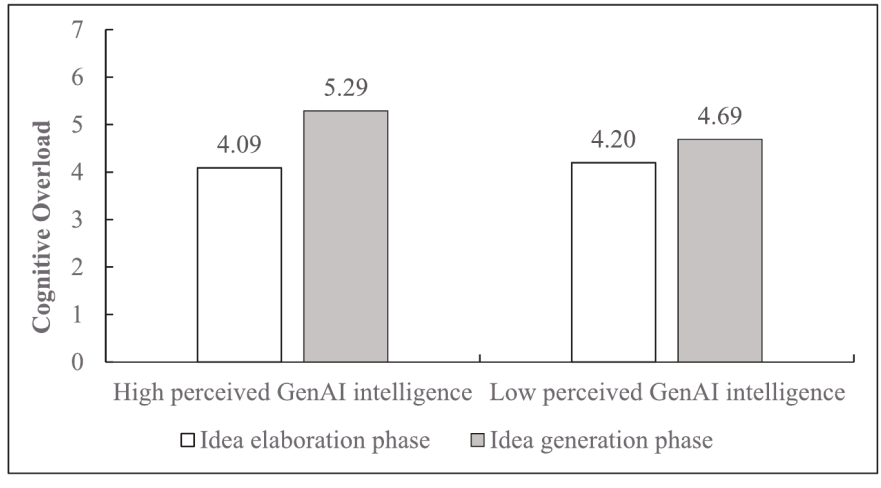


Fig. 2. The interactive effect between collaboration phases and perceived GenAI intelligence on cognitive overload in Study 3.

practice. In doing so, it provides a new lens through which the "black box" of human–AI collaboration in creativity can be clarified and directly responds to the following question from Belanche et al. (2024): "What are the implications of generative AI like ChatGPT on human cognitive and creative abilities?"

Second, our research contributes novel insights into the cognitive mechanisms through which human–GenAI collaboration influences creativity, revealing how its role shifts across creative phases. Although GenAI adoption has surged in creative industries (Bouschery et al., 2023), theoretical clarity regarding its psychological influence on human creativity remains limited. Most prior studies emphasize GenAI's instrumental or ideation-enhancing role (Girotra et al., 2023; Zhao et al., 2024) and often portray it as a passive assistant or substitutive tool (Raisch & Krakowski, 2021), neglecting its dynamic potential as a cognitive collaborator. Our study addresses this gap by identifying a phase-specific dual-pathway model: cognitive flexibility as the mechanism through which human–GenAI collaboration enhances novelty during idea generation and cognitive overload reduction as the mechanism supporting usefulness in elaboration. These findings bridge microlevel cognitive processes with macrolevel outcomes, advancing a situated view of GenAI-enabled creativity. Moreover, by demonstrating GenAI's adaptive transitions between exploratory and evaluative roles, we reconceptualize human–GenAI teaming as a coevolving system guided by collaborative goals, thereby enriching theories of AI complementarity (Raisch & Krakowski, 2021).

Third, our research clarifies the boundary conditions of human–GenAI collaboration by uncovering the asymmetric moderating roles of perceived GenAI intelligence and GenAI experience across creative phases. Prior research has focused primarily on the impacts of large language model capabilities and individual expertise on human–GenAI interaction outcomes (Dell'Acqua et al., 2023; Doshi & Hauser, 2024). In contrast, we redirect attention to factors intrinsic to the coevolutionary process: how creators' subjective perceptions of GenAI and their prior interaction histories influence collaboration outcomes. While existing studies often perceive GenAI as a uniform tool and regard technology as universally instrumental, we emphasize the role of human perceptions of GenAI as a critical factor influencing collaboration outcomes. Our findings reveal that high perceived GenAI intelligence fosters trust and improves usefulness during elaboration (Boussioux et al., 2024) while exerting limited influence on novelty during generation. These findings challenge the view of GenAI as a uniformly effective partner (Gretzel, 2025) and highlight the importance of aligning individual-level moderators with the cognitive demands of each phase. This asymmetry suggests that the trust rooted in perceived GenAI intelligence may support performance in evaluative tasks but also diminish critical engagement and undermine idea generation. In highlighting this tension, our study responds directly to growing concerns about the dark side of GenAI technologies (Belanche et al., 2024; Belk et al., 2023).

Finally, our research contributes to the creativity literature by introducing the human–GenAI interaction as a novel mechanism influencing individual creativity across phases. While existing research has emphasized primarily interpersonal factors, such as team dynamics and social networks, that shape idea development (Mannucci & Perry-Smith, 2022), our framework shifts the focus to technologically mediated collaboration. By conceptualizing GenAI as a cognitively responsive, nonhuman partner, we expand creativity theory to accommodate hybrid creative contexts involving algorithmic agents. Integrating human–GenAI collaboration into phase-specific models reframes the creative process as one that is jointly shaped by digital cognition and psychological fit. In doing so, our study offers theoretical tools for exploring how synthetic input influences creative outcomes and extends creativity research beyond its traditional human-centered boundaries.

7.2. Practical implications

Our findings offer actionable guidance for creative professionals and organizations by clarifying how to strategically align human–GenAI collaboration with the distinct cognitive goals of different creative phases. Across various creative contexts, this study underscores the importance of treating GenAI not as a generic assistant but as a phase-contingent collaborator whose value lies in supporting either novelty or usefulness, depending on cognitive demands.

For marketing professionals, engaging GenAI during the idea generation phase can spark novel campaign concepts by broadening conceptual associations and reducing self-censorship. For example, marketers tasked with naming seasonal campaigns, developing brand slogans, or crafting product narratives can leverage GenAI's lateral exploration capabilities to generate unexpected framing angles. In the elaboration phase, GenAI can help synthesize audience data (e.g., CRM systems or ad dashboards), simulate A/B testing outcomes, and flag executional risks to improve message precision and strategic alignment.

For design teams, especially in product or user experience (UX) settings, aligning GenAI use with creative phases can enhance ideation quality and workflow fluency. During early-stage brainstorming, GenAI can propose imaginative visual cues, design motifs, or usage scenarios, thereby stimulating team-level cognitive flexibility and challenging entrenched ways of thinking. In the elaboration phase, GenAI acts as a real-time reviewer, providing feedback to streamline functionality, reduce redundancies, and enhance feasibility without compromising originality. Tools such as GenAI-enabled plugins in Figma or prototyping platforms allow direct integration into existing design routines.

For digital product development teams operating under tight iteration cycles, tailoring GenAI deployment to phase-specific goals can facilitate smoother cross-functional collaboration. During discovery sprints, developers and product managers can harness GenAI as an "exploratory copilot" to expand feature ideation through diverse prompts. In subsequent build phases, GenAI can serve as a "refinement assistant" to clarify user stories, generate edge-case test scenarios, and simulate feedback loops. Prompt engineering tools and GenAI visualization dashboards (e.g., flow generators or persona simulations) can further increase collaboration fluency across roles.

In addition, our results highlight the need to calibrate creators' perceived GenAI intelligence on the basis of phase-specific requirements. During idea elaboration, reinforcing trust in GenAI's usefulness enhances cognitive bandwidth and evaluative confidence. However, in open-ended ideation, excessive reliance may constrain originality. Organizations should encourage employees to engage in critical thinking and exploratory use of GenAI during early-stage ideation and pursue deeper human–GenAI collaboration during idea elaboration. These approaches help foster both open-mindedness and discernment throughout the creative process. By applying these insights, creative teams and innovation managers can strategically allocate GenAI tasks, design context-sensitive workflows, and build collaborative cultures that recognize GenAI's shifting role in novelty versus usefulness. These implications provide not only operational clarity but also a pathway for expanding human–AI teamwork in hybrid creative environments.

7.3. Limitations and future directions

This study has several limitations that, in turn, point to avenues for future research. First, while we employed a rigorous multimethod design to enhance internal validity and contextual grounding, the generalizability of our findings may still be limited by the participant profiles and cultural context. Most participants were based in East Asia and engaged in individual creative tasks. Research suggests that creative professionals from different cultural backgrounds exhibit varying degrees of GenAI acceptance, usage behaviors, and creative outcomes (Belanche et al., 2020; Cao et al., 2023; Dabbous et al., 2022; Kim et al., 2022). Although our findings reveal that individuals prioritize novelty during idea generation and usefulness during elaboration, these goal-driven motivations may be culturally contingent. Future studies

could explore how cultural values (e.g., individualism vs. collectivism) shape human–GenAI collaboration patterns. For example, in collectivist cultures (e.g., China), researchers could examine how GenAI facilitates team-based ideation and goal convergence, whereas in individualist cultures (e.g., the U.S.), studies might focus on how GenAI enhances individual creative agency and self-expression.

Second, while this study emphasizes the benefits of human–GenAI collaboration for creativity, it does not fully address the dark side of GenAI. In recent years, scholars have called for closer attention to the "dark side" of GenAI use (Belanche et al., 2024; Belk et al., 2023), including its ethical, psychological, and cognitive risks. For example, early input from GenAI during idea generation may lead to premature fixation or reduced diversity (Yin et al., 2025), whereas delayed support during elaboration may discourage critical pruning or inflate low-potential ideas. These disruptions are likely to vary depending on the prompt structure, cognitive load, and task framing. Beyond phase-specific effects, long-term reliance on GenAI raises concerns about creative skill degradation and cognitive dependency (Macnamara et al., 2024). Sustained exposure may weaken users' tolerance for ambiguity and lead to the habitual outsourcing of creative judgment. To address these concerns, future research should investigate both the short-term dynamics and long-term trajectories of GenAI-enabled creativity and test interventions that help preserve critical autonomy in human–AI collaboration.

Third, while this study examines how perceived GenAI intelligence and prior GenAI experience moderate human–GenAI collaboration outcomes, future research should investigate a broader range of individual-level factors. Recent studies suggest that user characteristics, such as the need for social interaction, can significantly influence how individuals respond to AI systems (Belanche et al., 2021). By extending this logic to creative collaboration, future work could explore how personal traits (e.g., relational orientation, critical thinking) influence users' trust, reliance, and creative agency when they interact with GenAI across different phases (Lee et al., 2025). Overall, we call for future research to explore the boundary conditions under which GenAI functions as a constructive partner and when it may unintentionally constrain human creativity.

8. Conclusion

This study advances a novel phase-specific perspective on human–GenAI collaboration in creative work, with core findings revealing the dynamic dual role of GenAI across the creative process. It acts as a catalyst for novelty in the idea generation phase, where it activates cognitive flexibility to expand the breadth of creative exploration, and as an enabler of usefulness in the idea elaboration phase, where it mitigates cognitive overload to increase the depth of idea elaboration. This contribution breaks through the static view of GenAI as a monolithic tool and redefines it as a dynamic collaborator. Its creative amplification effect depends on cognitive alignment with the unique goals of each phase: novelty-driven in generation and usefulness-oriented in elaboration. Theoretically, this framework enriches emerging discussions on human–AI collaboration by demonstrating that effective creative collaboration is not a one-size-fits-all model but rather a context-sensitive interplay between technological capabilities and phase-specific cognitive demands.

Practically, our findings offer actionable guidelines for creative professionals, including marketers, UX designers, and product managers. Rather than treating GenAI as a universal assistant, creators should strategically deploy it on the basis of the creative phase. They can engage it as a divergent partner during early ideation to spark unconventional ideas and as a convergent advisor during elaboration to enhance feasibility and coherence. Training programs should focus on calibrating creators' ability to align GenAI with phase goals, thereby avoiding blind adoption that may stifle originality or hinder refinement.

For researchers, this study provides a generative framework to explore the nuanced dynamics of AI-enhanced creativity. Future work could further examine how factors such as domain specificity (e.g., visual design vs. writing), user expertise, and AI tool characteristics (e.g., prompt sensitivity, output diversity) moderate the phase-specific effects of GenAI. Additionally, investigating the long-term impact of phase-aligned collaboration on creative skill development could deepen our understanding of human–GenAI collaboration in creative domains. As GenAI reshapes creative industries, this research underscores that its transformative potential lies not in comprehensive replacement but in thoughtful, phase-aware integration. It leverages GenAI's generative strengths while preserving human judgment, intent, and contextual insight. This approach paves the way for a more synergistic future where human creativity and AI capabilities reinforce one another.

CRediT authorship contribution statement

Julie N. Y. Zhu: Conceptualization, Writing – review & editing. **Yanghao Zhu:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Lirong Long:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Shiyingzi Huang:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to polish the draft of the paper so as to improve the readability and language quality of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

We have no financial or non-financial conflicts of interest to disclose.

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Appendix A. Profiles of interview respondents

Response	Gender	Age	Education	Industry	Department
R1	Female	22	Bachelor	Online Fiction	Freelancer
R2	Male	27	Bachelor	Trade	Operations
R3	Female	28	Bachelor	Manufacturing	Design
R4	Male	26	Master	Media	Operations
R5	Female	25	Bachelor	Service	Operations
R6	Female	34	Master	IT technology	Operations
R7	Male	27	Bachelor	IT technology	Sale
R8	Female	28	Master	Education	Design
R9	Male	25	Bachelor	Construction	Design
R10	Female	24	Bachelor	IT technology	Operations
R11	Female	29	College	Service	Operations
R12	Female	24	Master	IT technology	Operations
R13	Female	33	Doctor	IT technology	R&D
R14	Female	26	Master	Education	Marketing
R15	Male	29	Bachelor	Media	Design
R16	Female	27	Bachelor	Service	Design
R17	Female	29	Master	IT technology	Marketing
R18	Male	32	Master	Media	Marketing
R19	Male	31	Bachelor	Manufacturing	Design
R20	Male	30	Bachelor	Manufacturing	Sale

Appendix B. Interview questions

1. To begin, could you briefly share your age, educational background, the industry you work in, your company's nature, your department, and a short description of your role?

2. When did you first start using GenAI in your work? What types of GenAI tools have you used?

3. What kinds of creative tasks do you typically complete with the help of GenAI? Could you walk me through a specific instance where you collaborated with GenAI to complete a creative project?

4. During which phases of the creative process do you usually engage with GenAI? How does your interaction with it typically unfold across those phases?

5. When collaborating with GenAI on creative tasks, what kinds of outcomes are you hoping to achieve?

6. In your view, what are the key ways GenAI impacts creative work when used in a collaborative setting?

7. How do you personally feel during the process of collaborating with GenAI to complete creative tasks?

Appendix C. Qualitative research coding framework in Study 1

Main Category	Sub-category	Initial Concept	Exemplary Quote
C1 the Human–GenAI Collaboration Phases	B1 Human–GenAI Collaboration in Idea Generation Phase	A1 Leveraging GenAI for Creative Inspiration	"Sometimes when I'm out of ideas, I let GenAI help brainstorm some plot points for inspiration." (1- 5)
		A2 Requesting Diverse Creative Ideas from GenAI	"I ask GenAI to innovate current marketing methods and generate low-cost, high-interaction solutions." (2- 3)
		A3 GenAI-assisted Creative Framework Generation	"I have GenAI generate story outlines; it provides multiple different narrative directions." (18- 3)
		A4 Using GenAI to Track Hot Topics	"For example, if I want to know how to tie in a popular TV show or movie for promotion, I have GenAI search the internet for relevant information to find a fresh promotional angle." (7- 2)
	B2 Human–GenAI Collaboration in Idea Elaboration Phase	A5 GenAI Refining Creative Details Based on Precise Instructions	"It helps me refine script details, such as designing dialogue, scenes, and emphasizing the lipstick's selling points." (11- 4)
		A6 Validating Idea Feasibility with GenAI	"In the idea elaboration phase, we use GenAI to evaluate feasibility and check for any oversights." (8- 3)
		A7 Using GenAI to Analyze Pros and Cons for Decision-making	"I have AI analyze the advantages and disadvantages of different plans and propose adding surprise elements, raffles, and other refinements." (2- 7)
		A8 Using GenAI to Develop Contingency Plans	"Just before an event starts, I use GenAI to anticipate potential risks, such as live stream lag or injuries during on-site experience tournaments, and get contingency plans." (15- 6)
C2 the Creative Goals of the Human–GenAI Collaboration	B3 Focus on Creative Novelty	A9 Selecting More Innovative Ideas	"When setting outlines, I focus on identifying hotspots and perspectives." (6- 11)
		A10 Seeking to Break Traditional Patterns	"I hope to break away from the traditional model of book publishing events and experiment with new formats and themes, incorporating current trending elements." (14- 8)
		A11 Pursuing Eye-catching Appeal	"During the proposal generation phase, I pay attention to novelty and consider how to grab more attention." (5- 12)
	B4 Focus on Creative Usefulness	A12 Emphasizing Logical Coherence	"When polishing details, I choose content that is logical and coherent, considering whether the ideas are practical." (1- 14)
		A13 Ensuring Feasibility	"I pay attention to whether it's realistic, for example, whether a raffle giveaway of clothes might lead to product complaints. If so, I'll choose to scrap it." (2- 11)

(*continued on next page*)

(*continued*)

Main Category	Sub-category	Initial Concept	Exemplary Quote
		A14 Assessing Activity Risks	"Considering potential issues, such as order volume, overbooking, or author signing capacity, and offering solutions to ensure smooth execution." (14- 11)
		A15 Balancing Cost and Benefit	"Using discounts instead of raffling clothes to control costs while boosting sales and purchase intent." (2- 13)
		A16 Integrating Creativity with Real Products	"In the next phase, I focus on how to seamlessly integrate the product, like designing scripts and arranging product placement. At this stage, usefulness becomes more important." (7- 15)
C3 Underlying Psychological Mechanisms	B5 Promoting Cognitive Flexibility	A17 Breaking Free from Traditional Thought Constraints	"GenAI designed kites with unconventional, fun imagery that defies traditional styles, creating innovative ideas that break away from the norm." (10- 18)
		A18 Providing Alternative Solutions	"GenAI offers a multitude of different event directions, like providing thirty to forty concepts for Women's Day events, expanding my imagination." (14- 13)
		A19 Generating Cross-Domain Integration Ideas	"It proposed the concept of a 'Tech Trendsetter Online Showcase,' combining product promotion with fashion outfits—an angle I had never considered before." (17- 12)
		A20 Expanding Creative Boundaries	"It proposed unexpected ideas like designing weddings as 'travel adventures' with tasks for guests acting as 'travel companions'." (16- 17)
	B6 Managing Cognitive Load	A21 Alleviating Anxiety and Fear of Difficulty	"GenAI alleviates my reluctance to take on new topics and offers psychological support. When leaders or partners need proposals within a short time, GenAI helps me gain more time and relieve anxiety." (14- 18)
		A22 Enhancing Plan Feasibility Verification	"GenAI helps me analyze the quality of my plan, reducing trial-and-error costs." (4- 20)
		A23 Mitigating Activity Risks	"GenAI proactively provides potential issues and precautions for events, such as insufficient order volume, overbooking, or the author being overwhelmed, along with solutions." (14- 17)
		A24 Reducing Structural Work Pressure	"GenAI can handle these structured tasks, reducing my cognitive pressure." (6- 16)
		A25 Reducing the Stress of Colleague Coordination and Consultation	"GenAI is like a reliable colleague who always has my back, never lets me down, and is always ready to help. This is far better than dealing with real humans in the workplace. In a professional setting, no one is always available to meet your every request, no matter how demanding it may be. GenAI never gives negative feedback, whereas when you try to work out a plan with a colleague, differences in perspectives can easily stall progress and make it hard to move forward." (14- 23)

Note: Due to the length of the article, exemplary quote shows only part of the data. (X-Y) means the Y-th coded statement in the X-th sample.

Appendix D. Tasks in Study 2

Task A/B/C/D is a creative design task focused on a mouse/milk/headphone/bicycle product. Imagine you are a product manager at a technology company / dairy company that plans to launch a new type of mouse / milk / headphone / bicycle. The company is planning to make this product more innovative (novel and useful) compared to existing alternatives in the market. You are invited to contribute to the creative planning process for this product. Please write a creative proposal of at least 100 words. The task will be completed in 20 min and consists of two phases.

Appendix E. Manipulation material in Study 2a

For the condition collaborating with GenAI in the idea generation phase (Coded as 1), the following instructions are given "First, please engage in broad thinking to conceive what you believe to be the most creative initial ideas. The idea generation phase will last for 10 min. After 10 min, please fill in your preliminary creative ideas in the text box below. Please note: During this phase, you are allowed to collaborate with the GenAI tool. Subsequently, please elaborate and perfect the preliminary ideas generated in the previous phase. The idea elaboration phase will last for 10 min. After 10 min, please fill in your final creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI."

For the condition working independently in the idea generation phase (Coded as 0), the following instructions are given "First, please engage in broad thinking to conceive what you believe to be the most creative initial ideas. The idea generation phase will last for 10 min. After 10 min, please fill in your preliminary creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI. Subsequently, please elaborate and perfect the preliminary ideas generated in the previous phase. The idea elaboration phase will last for 10 min. After 10 min, please fill in your final creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI."

Appendix F. Manipulation material in Study 2b

For the condition collaborating with GenAI in the idea elaboration phase (Coded as 1), the following instructions are given "First, please engage in broad thinking to conceive what you believe to be the most creative initial ideas. The idea generation phase will last for 10 min. After 10 min, please fill in your preliminary creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI. Subsequently, please elaborate and perfect the preliminary ideas generated in the previous phase. The idea elaboration phase will last for 10 min. After 10 min, please fill in your final creative ideas in the text box below. Please note: During this phase, you are allowed to collaborate with the GenAI tool."

For the condition working independently in the idea elaboration phase (Coded as 0), the following instructions are given "First, please engage in broad thinking to conceive what you believe to be the most creative initial ideas. The idea generation phase will last for 10 min. After 10 min, please

fill in your preliminary creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI. Subsequently, please elaborate and perfect the preliminary ideas generated in the previous phase. The idea elaboration phase will last for 10 min. After 10 min, please fill in your final creative ideas in the text box below. Please note: During this process, do not use search engines or tools like GenAI."

Appendix G. Task in Study 3

Please read the following content and complete the creative design task for the phone case proposal as required. You work as a product manager for Apple's iPhone division. The company plans to launch a new phone case, and you need to create advertising copy for it. Below are the product details:

Product Name: iPhone Leather Flip Case

Features & Highlights: (1) Material: Premium retro PU leather with soft TPU for unmatched durability. Easy to clean with a damp cloth. (2) Design: Features a built-in stand and a luxury cardholder pocket. Provides 360° protection with raised edges to safeguard the camera and screen. Scratch-resistant, anti-fingerprint finish. Internal anti-slip lining offers shock absorption. (3) Functionality: Built-in stand enables hands-free video watching and multitasking. (4) Compatibility: Precision fit for multiple iPhone models. Full access to ports, buttons, and sensors. (5) Price: Original ¥60; Promo Price: ¥30 (50 % off) with free shipping.

Task Requirements: Prioritize novelty and practicality in your creative proposal (200 + words). Complete the task in 20 min, divided into two phases: "idea generation phase" (10 mins) and "idea elaboration phase" (10 mins).

Appendix H. Manipulation material in Study 3

Perceiving Low Intelligence in Generative AI:

First, please read the following news article outlining the current capabilities of generative AI. "Generative AI Intelligence Remains in Its Infancy": In recent years, generative artificial intelligence (GenAI), exemplified by tools like ChatGPT, has demonstrated the ability to produce content and ideas, including conversations, stories, images, videos, and music. However, in creative tasks, GenAI's capabilities remain mechanical and formulaic, falling short of serving as robust collaborative partners for humans. GenAI struggles to sensitively interpret nuanced human needs or respond dynamically to personalized demands. Its comprehension of complex tasks is limited, often yielding shallow or impractical creative outputs. It lacks precision in grasping market dynamics, resulting in overly generalized recommendations and subpar optimization solutions. Overall, GenAI's intelligence remains rudimentary and underdeveloped.

Perceiving High Intelligence in Generative AI:

First, please read the following news article outlining the current capabilities of generative AI. "Generative AI Displays Remarkable Intelligence": In recent years, generative artificial intelligence (GenAI), exemplified by tools like ChatGPT, has demonstrated groundbreaking abilities to produce creative content and ideas, including conversations, stories, images, videos, and music. In creative domains, GenAI now exhibits human-like adaptability, emerging as a powerful ally for human innovation. Equipped with keen sensitivity to interpret nuanced human needs, it actively responds to personalized requests. GenAI rapidly processes complex tasks, generating imaginative and impactful ideas with profound market insights. Its recommendations reflect professional-grade expertise, and it delivers exceptional efficiency in refining solutions. Collectively, GenAI has evolved into a highly sophisticated and intelligent system.

Appendix I. Adapted items about core variables

Variables	Items	Source
Cognitive Flexibility	When dealing with the creative task,…1. I consider alternatives. 2. I can find workable solutions to seemingly unsolvable problems. 3. I have the self-confidence necessary to try different ways of solutions.	Kundro, (2023)
Cognitive Overload	When dealing with the creative task,…1. The creative task is mentally demanding. 2. The pace of the creative task is hurried and rushed. 3. I have to work hard to accomplish my level of performance.	Sung et al. (2017)
Perceived GenAI Intelligence	1. I think GenAI is conscious. 2. I think GenAI is humanoid 3. I think GenAI is intelligent. 4. I think GenAI is sensitive. 5. I think GenAI is actively responsive.	Chen et al. (2023)
GenAI Experience	1. I used GenAI to carry out most of my job functions. 2. I spent most of the time working with GenAI. 3. I worked with GenAI in making major work decisions.	Man Tang et al. (2022)

References

Acar, S., & Runco, M. A. (2019). Divergent thinking: New methods, recent research, and extended theory. *Psychology of Aesthetics, Creativity, and the Arts, 13*(2), 153–158.

Amabile, T. M. (1983). The social psychology of creativity: A componential conceptualization. *Journal of Personality and Social Psychology, 45*(2), 357–376.

Amabile, T. M., Conti, R., Coon, H., Lazenby, J., & Herron, M. (1996). Assessing the work environment for creativity. *Academy of Management Journal, 39*(5), 1154–1184.

Baas, M., De Dreu, C. K., & Nijstad, B. A. (2011). When prevention promotes creativity: The role of mood, regulatory focus, and regulatory closure. *Journal of Personality and Social Psychology, 100*(5), 794–809.

Beghetto, R. A., & Kaufman, J. C. (2007). Toward a broader conception of creativity: A case for "mini-c" creativity. *Psychology of Aesthetics, Creativity, and the Arts, 1*(2), 73–79.

Belanche, D., Belk, R. W., Casaló, L. V., & Flavián, C. (2024). The dark side of artificial intelligence in services. *The Service Industries Journal, 44*(3-4), 149–172.

Belanche, D., Casaló, L. V., Flavian, C., & Schepers, J. (2020). Service robot implementation: A theoretical framework and research agenda. *The Service Industries Journal, 40*(3-4), 203–225.

Belanche, D., Casaló, L. V., Schepers, J., & Flavián, C. (2021). Examining the effects of robots' physical appearance, warmth, and competence in frontline services: The Humanness-Value-Loyalty model. *Psychology Marketing, 38*(12), 2357–2376.

Belk, R. W., Belanche, D., & Flavian, C. (2023). Key concepts in artificial intelligence and technologies 4.0 in services. *Service Business, 17*(1), 1–9.

Benbya, H., Strich, F., & Tamm, T. (2024). Navigating generative artificial intelligence promises and perils for knowledge and creative work. *Journal of the Association for Information Systems, 25*(1), 23–36.

Bouschery, S. G., Blazevic, V., & Piller, F. T. (2023). AI-augmented creativity: Overcoming the productivity loss in brainstorming groups. In *In Academy of Management Proceedings, 2023* p. 11938). Briarcliff Manor, NY 10510: Academy of Management.

Boussioux, L., Lane, J. N., Zhang, M., Jacimovic, V., & Lakhani, K. R. (2024). The crowdless future? Generative AI and creative problem-solving. *Organization Science, 35*(5), 1589–1607.

Boyacı, T., Canyakmaz, C., & de Véricourt, F. (2024). Human and machine: the impact of machine input on decision making under cognitive limitations. *Management Science, 70*(2), 1258–1275.

Burmeister, A., Alterman, V., Fasbender, U., & Wang, M. (2022). Too much to know? The cognitive demands of daily knowledge seeking and the buffering role of coworker contact quality. *Journal of Applied Psychology, 107*(8), 1303–1322.

Cao, Y., Zhou, L., Lee, S., Cabello, L., Chen, M., & Hershcovich, D. (2023). Assessing cross-cultural alignment between ChatGPT and human societies: An empirical study. *In Proceedings of the First Workshop on Cross-Cultural Considerations in NLP (C3NLP)*, 53–67.

Chen, A., Xiang, M., Wang, M., & Lu, Y. (2023). Harmony in intelligent hybrid teams: The influence of the intellectual ability of artificial intelligence on human members' reactions. *information technology &. People, 36*(7), 2826–2846.

Chen, Z., & Chan, J. (2024). Large language model in creative work: The role of collaboration modality and user expertise. *Management Science, 70*(12), 9101–9117.

Cheng, Q., Sun, T., Zhang, W., Wang, S., Liu, X., Zhang, M., … & Qiu, X.. (2023). Evaluating hallucinations in Chinese large language models. https://doi.org/10.48550/ARXIV.2310.03368.

Chowdhury, S., Budhwar, P., Dey, P. K., Joel-Edgar, S., & Abadie, A. (2022). AI-employee collaboration and business performance: Integrating knowledge-based view, socio-technical systems and organisational socialisation framework. *Journal of Business Research, 144*, 31–49.

Chowdhury, S., Dey, P., Joel-Edgar, S., et al. (2023). Unlocking the value of artificial intelligence in human resource management through AI capability framework. *Human Resource Management Review, 33*(1), Article 100899.

Criscuolo, P., Salter, A., & Ter Wal, A. L. J. (2014). Going underground: Bootlegging and individual innovative performance. *Organization Science, 25*(5), 1287–1305.

Cromwell, J.R., Harvey, J., Haase, J., & Gardner, H.K. (2023). Discovering Where ChatGPT can create value for your company. Harvard Business Review. https://hbr.org/2023/06/discovering-where-chatgpt-can-create-value-for-your-company.

da Silva Cezar, B. G., & Maçada, A. C. G. (2023). Cognitive overload, anxiety, cognitive fatigue, avoidance behavior and data literacy in big data environments. *Information Processing Management, 60*(6), Article 103482.

Dabbous, A., Aoun Barakat, K., & Merhej Sayegh, M. (2022). Enabling organizational use of artificial intelligence: An employee perspective. *Journal of Asia Business Studies, 16*(2), 245–266.

Dane, E. (2010). Reconsidering the trade-off between expertise and flexibility: A cognitive entrenchment perspective. *Academy of Management Review, 35*(4), 579–603.

Dell'Acqua, F., McFowland, E., Mollick, E.R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., … & Lakhani, K.R. (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. Harvard Business School Technology & Operations Mgt. Unit Working Paper, 24-013.

Dechaume, M., Mercier, M., Feybesse, C., Lubart, T., Chouvelon, G., Kermarrec, S., & Tordjman, S. (2024). Is intelligence necessary and sufficient for creativity? An analysis of convergent and divergent thinking. *Learning and Individual Differences, 116*, 102575.

Dinger, F. C., Dickhäuser, O., Spinath, B., & Steinmayr, R. (2013). Antecedents and consequences of students' achievement goals: A mediation analysis. *Learning and Individual Differences, 28*, 90–101.

Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances, 10*(28), eadn5290.

Dunn, J., Ruedy, N. E., & Schweitzer, M. E. (2012). It hurts both ways: How social comparisons harm affective and cognitive trust. *Organizational Behavior and Human Decision Processes, 117*(1), 2–14.

Duong, C. D., Vu, T. N., & Ngo, T. V. N. (2023). Applying a modified technology acceptance model to explain higher education students' usage of ChatGPT: A serial multiple mediation model with knowledge sharing as a moderator. *The International Journal of Management Education, 21*(3), Article 100883.

Dweck, C. S. (1986). Motivational processes affecting learning. *American Psychologist, 41*(10), 1040–1048.

Dwivedi, Y. K., Hughes, L., Bhadeshia, H. K., Ananiadou, S., Cohn, A. G., Cole, J. M., & Wang, X. (2024). Artificial intelligence (AI) futures: India-UK collaborations emerging from the 4th royal society yusuf hamied workshop. *International Journal of Information Management, 76*, Article 102725.

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., & Wright, R. (2023). Opinion paper: "So what if ChatGPT wrote it?" multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, Article 102642.

Elliot, A., & Church, M. A. (1997). A hierarchical model of approach and avoidance achievement motivation. *Journal of Personality and Social Psychology, 72*(1), 218–232.

Elliot, A. J., Murayama, K., & Pekrun, R. (2011). A 3*2 achievement goal model. *Journal of Educational psychology, 103*(3), 632–648.

Epstein, Z., Hertzmann, A., Investigators of Human Creativity, Akten, M., Farid, H., Fjeld, J., & Smith, A. (2023). Art and the science of generative AI. *Science, 380*(6650), 1110–1111.

Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G* power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191.

Finke, R.A., Ward, T.B., & Smith, S.M. (1996). *Creative cognition: Theory, research, and applications*. MIT press.

Flavián, C., Belk, R. W., Belanche, D., & Casaló, L. V. (2024). Automated social presence in AI: Avoiding consumer psychological tensions to improve service value. *Journal of Business Research, 175*, Article 114545.

Flavián, C., Pérenz-Rueda, A., Belanche, D., & Casaló, L. V. (2022). Intention to use analytical artificial intelligence (AI) in services–the effect of technology readiness and awareness. *Journal of Service Management, 33*(2), 293–320.

Forster, J., & Dannenberg, L. (2010). GLOMOsys: A systems account of global versus local processing. *Psychological Inquiry, 21*(3), 175–197.

Fui-Hoon Nah, F., Zheng, R., Cai, J., Siau, K., & Chen, L. (2023). Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. *journal of information technology case and application. Research, 25*(3), 277–304.

GAO, W., LI, Y., HUANG, Y., & YUAN, J. (2025). The effect of goal and situation task-switching training on emotion regulation flexibility and its mechanisms. *Advances in Psychological Science, 33*(2), 202–211.

Gehman, J., Glaser, V. L., & Merritt, P. (2025). An assemblage perspective on hybrid agency: A commentary on raisch and Fomina's "Combining human and artificial Intelligence". *Academy of Management Review, 50*(2), 477–479.

Gilhooly, K. J., & Murphy, P. (2005). Differentiating insight from non-insight problems. *Thinking Reasoning, 11*(3), 279–302.

Girotra, K., Meincke, L., Terwiesch, C., & Ulrich, K.T. (2023). Ideas are dimes a dozen: Large language models for idea generation in innovation. https://doi.org/10.2139/ssrn.4526071.

Grand, J. A., Braun, M. T., Kuljanin, G., Kozlowski, S. W. J., & Chao, G. T. (2016). The dynamics of team cognition: A process-oriented theory of knowledge emergence in teams. *Journal of Applied Psychology, 101*(10), 1353–1385.

Gretzel, U. (2025). The many questions concerning technology: Implications for tourism, hospitality, events and leisure research. *Journal of Hospitality and Tourism Management, 62*, 232–236.

Guest, G., Bunce, A., & Johnson, L. (2006). How many interviews are enough? An experiment with data saturation and variability. *Field Methods, 18*(1), 59–82.

Guilford, J. P. (1967). Creativity: Yesterday, today and tomorrow. *The Journal of Creative Behavior, 1*(1), 3–14.

Han, T., Xu, G., & Lu, W. (2025). Examining the effects of different types of achievement goal orientation on undergraduate students' engagement in distance learning: The mediating effect of self-efficacy. *Behavioral Sciences, 15*(1), 39. https://doi.org/10.3390/bs15010039

Harrison, S. H., & Rouse, E. D. (2015). An inductive study of feedback interactions over the course of creative projects. *Academy of Management Journal, 58*(2), 375–404.

Hart, S. G. (1988). Development of a multi-dimensional workload rating scale: Results of empirical and theoretical research. *Human Mental Workload, 1*, 39–183.

Hayes, A.F. (2017). Introduction to mediation, moderation, and conditional process analysis: A regression-based approach. New York: Guildford Press.

Jeong, I., Gong, Y., & Zhong, B. (2023). Does an employee-experienced crisis help or hinder creativity? An integration of threat-rigidity and implicit theories. *Journal of Management, 49*(4), 1394–1429.

Jia, N., Luo, X., Fang, Z., & Liao, C. (2024). When and how artificial intelligence augments employee creativity. *Academy of Management Journal, 67*(1), 5–32.

Keem, S., Shalley, C. E., Kim, E., & Jeong, I. (2018). Are creative individuals bad apples? A dual pathway model of unethical behavior. *Journal of Applied Psychology, 103*(4), 416–431.

Kim, J. H., Jung, H. S., Park, M. H., Lee, S. H., Lee, H., Kim, Y., & Nan, D. (2022). Exploring cultural differences of public perception of artificial intelligence via big data approach (June). *In International conference on human-computer interaction* (pp. 427–432). Cham: Springer International Publishing (June).

Kong, H., Yin, Z., Baruch, Y., & Yuan, Y. (2023). The impact of trust in AI on career sustainability: The role of employee–AI collaboration and protean career orientation. *Journal of Vocational Behavior, 146*, Article 103928.

Kshetri, N., Dwivedi, Y. K., Davenport, T. H., & Panteli, N. (2024). Generative artificial intelligence in marketing: Applications, opportunities, challenges, and research agenda. *International Journal of Information Management, 75*, Article 102716.

Kundro, T. G. (2023). The benefits and burdens of work moralization on creativity. *Academy of Management Journal, 66*(4), 1183–1208.

Lalot, F., & Bertram, A. M. (2025). When the bot walks the talk: Investigating the foundations of trust in an artificial intelligence (AI) chatbot. *Journal of Experimental Psychology: General, 154*(2), 533–551.

Lee, B. C., & Chung, J. (2024). An empirical investigation of the impact of ChatGPT on creativity. *nature human. Behaviour, 8*(10), 1906–1914.

Lee, H. P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers (April) *In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–22.

Lin, Z. (2024). How to write effective prompts for large language models. *Nature Human Behaviour, 8*(4), 611–615.

Liu, Y., Li, Y., Song, K., & Chu, F. (2024). The two faces of artificial intelligence (AI): analyzing how AI usage shapes employee behaviors in the hospitality industry. *International Journal of Hospitality Management, 122*, Article 103875.

Macnamara, B. N., Berber, I., Çavuşoğlu, M. C., Krupinski, E. A., Nallapareddy, N., Nelson, N. E., & Ray, S. (2024). Does using artificial intelligence assistance accelerate skill decay and hinder skill development without performers' awareness. *Cognitive Research Principles and Implications, 9*(1), 46. https://doi.org/10.1186/s41235-024-00572-8

Madjar, N., Greenberg, E., & Chen, Z. (2011). Factors for radical creativity, incremental creativity, and routine, noncreative performance. *Journal of Applied Psychology, 96*(4), 730–743.

Mainemelis, C. (2010). Stealing fire: Creative deviance in the evolution of new ideas. *Academy of Management Review, 35*(4), 558–578.

Malik, A., Budhwar, P., Mohan, H., & NR, S. (2023). Employee experience–the missing link for engaging employees: Insights from an MNE's AI-based HR ecosystem. *Human Resource Management, 62*(1), 97–115.

Man Tang, P., Koopman, J., McClean, S. T., Zhang, J. H., Li, C. H., De Cremer, D., & Ng, C. T. S. (2022). When conscientious employees meet intelligent machines: an integrative approach inspired by complementarity theory and role theory. *Academy of Management Journal, 65*(3), 1019–1054.

Maner, J. K., DeWall, C. N., Baumeister, R. F., & Schaller, M. (2007). Does social exclusion motivate interpersonal reconnection? Resolving the " porcupine problem. *Journal of Personality and Social Psychology, 92*(1), 42–55.

Mannucci, P. V., & Perry-Smith, J. E. (2022). "Who are you going to call?" network activation in creative idea generation and elaboration. *Academy of Management Journal, 65*(4), 1192–1217.

Martin, M. M., & Rubin, R. B. (1995). A new measure of cognitive flexibility. *Psychological Reports, 76*(2), 623–626.

Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods, 17*(3), 437–455.

Miron-Spektor, E., Vashdi, D. R., & Gopher, H. (2022). Bright sparks and enquiring minds: Differential effects of goal orientation on the creativity trajectory. *Journal of Applied Psychology, 107*(2), 310–318.

Montoya, A. K., & Hayes, A. F. (2017). Two-condition within-participant statistical mediation analysis: A path-analytic framework. *Psychological Methods, 22*(1), 6–27.

Mueller, J. S., & Kamdar, D. (2011). Why seeking help from teammates is a blessing and a curse: A theory of help seeking and individual creativity in team contexts. *Journal of Applied Psychology, 96*(2), 263–276.

Mumford, M. D. (2000). Managing creative people: Strategies and tactics for innovation. *Human Resource Management Review, 10*(3), 313–351.

Mumford, M. D., Hester, K. S., & Robledo, I. C. (2012). Creativity in organizations: Importance and approaches. *In Handbook of organizational creativity* (pp. 3–16). Academic Press.

Nedungadi, P., Ramesh, M., Govindaraju, V., Rao, B., Berbeglia, P., & Raman, R. (2024). Emerging leaders or persistent gaps? Generative AI research May foster women in STEM. *International Journal of Information Management, 77*, Article 102785.

Nicholls, J. G. (1984). Achievement motivation: Conceptions of ability, subjective experience, task choice, and performance. *Psychological Review, 91*(3), 328–346.

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science, 381*(6654), 187–192.

Perry-Smith, J. E., & Mannucci, P. V. (2017). From creativity to innovation: The social network drivers of the four phases of the idea journey. *Academy of Management Review, 42*(1), 53–79.

Pezzulo, G., Parr, T., Cisek, P., Clark, A., & Friston, K. (2024). Generating meaning: Active inference and the scope and limits of passive AI. *Trends in Cognitive Sciences, 28*(2), 97–112.

Raisch, S., & Fomina, K. (2025). Combining human and artificial intelligence: hybrid problem-solving in organizations. *Academy of Management Review, 50*(2), 441–464.

Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: the automation–augmentation paradox. *Academy of Management Review, 46*(1), 192–210.

Rhue, L. (2023). The anchoring effect, algorithmic fairness, and the limits of information transparency for emotion artificial intelligence. *Information Systems Research, 35*(3), 1479–1496.

Riemer, K., & Peter, S. (2024). Conceptualizing generative AI as style engines: Application archetypes and implications. *International Journal of Information Management, 79*, Article 102824.

Runco, M. A., & Jaeger, G. J. (2012). The standard definition of creativity. *Creativity Research Journal, 24*(1), 92–96.

Sassenberg, K., Winter, K., Becker, D., Ditrich, L., Scholl, A., & Moskowitz, G. B. (2022). Flexibility mindsets: reducing biases that result from spontaneous processing. *European Review of Social Psychology, 33*(1), 171–213.

Seo, Y. W., Chae, S. W., & Lee, K. C. (2015). The impact of absorptive capacity, exploration, and exploitation on individual creativity: Moderating effect of subjective well-being. *Computers in Human Behavior, 42*, 68–82.

Shao, Y., Huang, C., Song, Y., Wang, M., Song, Y. H., & Shao, R. (2024). Using augmentation-based AI tool at work: A daily investigation of learning-based benefit and challenge. *Journal of Management*. https://doi.org/10.1177/01492063241266503

Shao, Z., Li, X., & Wang, Q. (2022). From ambidextrous learning to digital creativity: an integrative theoretical framework. *Information Systems Journal, 32*(3), 544–572.

Simmons, A. L., & Ren, R. (2009). The influence of goal orientation and risk on creativity. *Creativity Research Journal, 21*(4), 400–408.

Siemon, D. (2022). Let the computer evaluate your idea: Evaluation apprehension in human-computer collaboration. *Behaviour Information Technology, 42*(5), 459–477.

Singla, A., Sukharevsky, A., Yee, L., Chui, M., & Hall, B. (2024). *The state of AI in early 2024: Gen AI adoption spikes and starts to generate value*. Mckinsey & Company.

Sohn, K., & Kwon, O. (2020). Technology acceptance theories and factors influencing artificial Intelligence-based intelligent products. *Telematics and Informatics, 47*, Article 101324.

Sung, S. Y., Antefelt, A., & Choi, J. N. (2017). Dual effects of job complexity on proactive and responsive creativity: Moderating role of employee ambiguity tolerance. *Group Organization Management, 42*(3), 388–418.

Tang, C., & Naumann, S. E. (2016). The impact of three kinds of identity on research and development employees' incremental and radical creativity. *Thinking Skills and Creativity, 21*, 123–131.

Tierney, P., & Farmer, S. M. (2002). Creative self-efficacy: Its potential antecedents and relationship to creative performance. *Academy of Management Journal, 45*(6), 1137–1148.

Wadinambiarachchi, S., Kelly, R. M., Pareek, S., Zhou, Q., & Velloso, E. (2024). The effects of generative ai on design fixation and divergent thinking. *In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 1–18.

Wang, H., Zhang, H., Chen, Z., et al. (2022). Influence of artificial intelligence and robotics awareness on employee creativity in the hotel industry. *Frontiers in Psychology, 13*, Article 834160.

Warner, R. M., & Sugarman, D. B. (1986). Attributions of personality based on physical appearance, speech, and handwriting. *Journal of Personality and Social Psychology, 50*(4), 792–799.

White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., … & Schmidt, D.C. (2023). A prompt pattern catalog to enhance prompt engineering with ChatGPT. https://doi.org/10.48550/arXiv.2302.11382.

Wirtz, J., Patterson, P. G., Kunz, W. H., Gruber, T., Lu, V. N., Paluch, S., & Martins, A. (2018). Brave new world: Service robots in the frontline. *Journal of Service Management, 29*(5), 907–931.

Yam, K. C., Bigman, Y. E., Tang, P. M., Ilies, R., De Cremer, D., Soh, H., & Gray, K. (2021). Robots at work: People prefer—and forgive—service robots with perceived feelings. *Journal of Applied Psychology, 106*(10), 1557–1577.

Yang, J., Mossholder, K. W., & Peng, T. K. (2009). Supervisory procedural justice effects: The mediating roles of cognitive and affective trust. *The Leadership Quarterly, 20*(2), 143–154.

Yeo, G. B., Celestine, N. A., Parker, S. K., To, M. L., & Hirst, G. (2024). A neurocognitive framework of attention and creativity: Maximizing usefulness and novelty via directed and undirected pathways. *Journal of Organizational Behavior, 45*(6), 912–934.

Yin, J., Ngiam, K. Y., Tan, S. S. L., & Teo, H. H. (2025). Designing AI-based work processes: How the timing of ai advice affects diagnostic decision making. *Management Science*. https://doi.org/10.1287/mnsc.2022.01454

Yin, M., Jiang, S., & Niu, X. (2024). Can AI really help? The double-edged sword effect of AI assistant on employees' innovation behavior. *Computers in Human Behavior, 150*, Article 107987.

Zahra, S. A., Neubaum, D. O., & Hayton, J. (2020). What do we know about knowledge integration: Fusing micro- and macro-organizational perspectives. *Academy of Management Annals, 14*(1), 160–194.

Zamfirescu-Pereira, J. D., Wong, R. Y., Hartmann, B., & Yang, Q. (2023). Why johnny can't prompt: How non-AI experts try (and fail) to design LLM prompts. *In Proceedings of the 2023 CHI Conference on Human factors in Computing Systems*, 1–21.

Zhang, T., Gino, F., & Margolis, J. D. (2018). Does "could" lead to good? On the road to moral insight. *Academy of Management Journal, 61*(3), 857–895.

Zhang, Y., Zhang, J., Gu, J., & Tse, H. H. (2022). Employee radical creativity: The roles of supervisor autonomy support and employee intrinsic work goal orientation. *Innovation, 24*(2), 272–289.

Zhao, W., Wang, W., & Viswanathan, S. (2024). Spillover effects of generative AI on human-generated content creation: Evidence from a crowd-sourcing design Platform. https://doi.org/10.2139/ssrn.4693181.

Zhou, E., & Lee, D. (2024). Generative artificial intelligence, human creativity, and art. *PNAS Nexus, 3*(3), 1–8. https://doi.org/10.1093/pnasnexus/pgae052